

**Evaluating the Relationship between Magnetic Damper Distance and Damping Ratio in an
Oscillating Spring-Mass System**

Laboratory Report

March 13, 2026

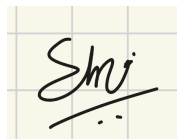
Patrick Jacobs

A handwritten signature in black ink, appearing to read "Patrick Jacobs", with a stylized flourish at the end.

Steven Ramsey

A handwritten signature in black ink, appearing to read "Steven Ramsey", with a stylized flourish at the end.

Shrish Rao

A handwritten signature in black ink, appearing to read "Shrish Rao", with a stylized flourish at the end, set against a light yellow grid background.

Introduction + Hypothesis

When a magnet moves across a material that is conductive, its magnetic field creates currents in the material that induce a magnetic field opposing the motion of the magnet. These currents are known as eddy currents, and their strength is dependent on that of the magnetic field. By controlling the height of the magnet from the conductive surface, one can control the magnetic field's strength, the eddy currents, and the strength of the opposing force, $F \propto B^2$ (Sodono & Bae, 2004). This opposing force can be used to damp a mass-spring system by reliably opposing its oscillation, and the damping ratio of the system can then be modeled as a function of the magnetic damper's height (Nagaya et. al., 1984). Typically, the relationship between a magnetic field's strength and displacement from the field's center can be modeled as $B \propto \frac{1}{r^3}$, based on the equation for the strength of a magnetic dipole's field. However, this relationship is found to be variable, and at close distances may be closer approximated by $B \propto \frac{1}{r^2}$ or $\frac{1}{r}$. This is due to the magnet behaving like a monopole instead of a dipole at near distances, and at a very near distance similarly to an infinitely extended wire. The transitional region in which this relationship changes before its continuous behavior as a dipole translates to an inconsistent relationship between damping ratio and height, under which none of them may accurately predict the damping ratio.

By attaching a magnetic damper to the mass in a spring-mass-spring system, oscillation of the system when displaced a standard distance from equilibrium can be used to find the damping ratio, ζ . A collection of damping ratios for differing heights of the magnetic damper can then be collected and plotted against each other and the relationship between ζ and height can be found. Our expectation was that this relationship can be accurately modeled with an exponential

approximation that approaches 1 (critically damped) as the magnetic damper approaches the track, and tends toward 0 (undamped) at increasingly further distances. To validate this hypothesis, a theoretical model of damping ratio vs. height for an altered spring-mass system will be created from initial trials and then compared against experimental measurements of the altered system to determine if the model is reasonably accurate.

Procedure + Methods

The initialization process for the experiment included finding the mass of the cart and masses (with and without the magnetic damping apparatus attached). The cart was attached to a spring on each side and placed on a track—each spring’s other side (non-adjacent to the cart) was fixed at a point. Each spring constant k was found by fixing one end of the spring, applying a fixed mass of 50 grams to the other, allowing it to extend due to gravity, and measuring the displacement (d). Each k is equal to the applied force divided by displacement, $k = \frac{mg}{d}$. Then, since the spring-mass system is in parallel, the equivalent spring constant is, $k_{eq} = k_1 + k_2$. The equilibrium (resting) position of the cart in the system was noted, and the width of the notecards which will be used to measure the varying heights of the magnetic damper.

Table 1

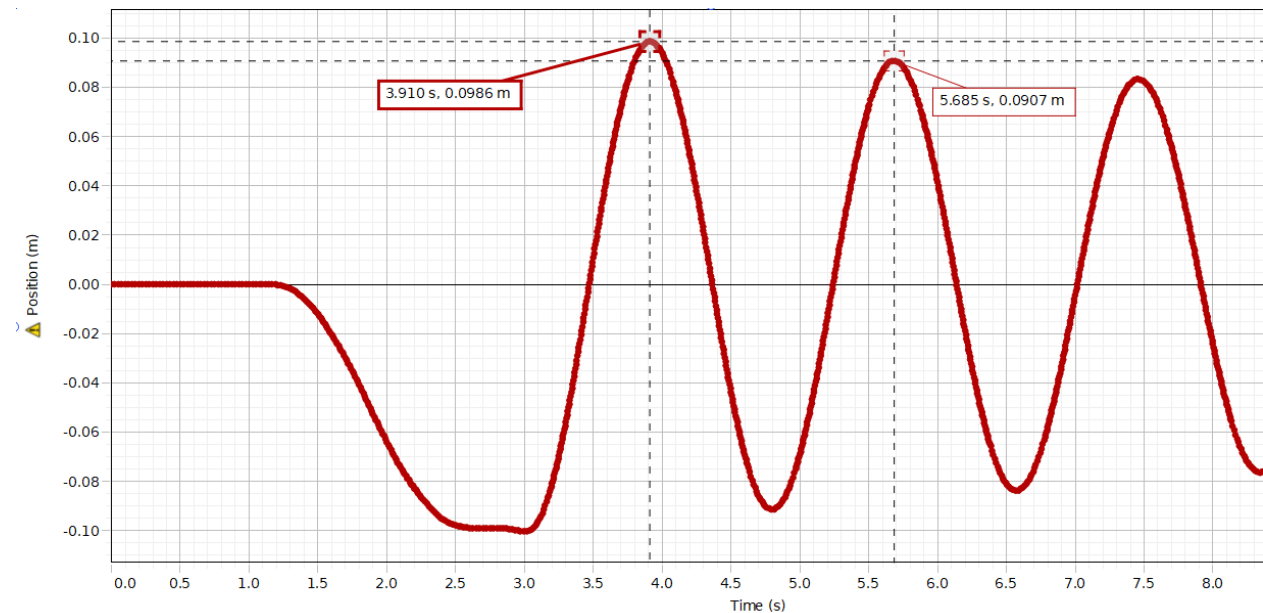
Spring analysis for experiment setup

Spring	Applied Mass (kg)	Displacement (m)	Spring Constant, k (N/m)
k_1	0.050	0.14	3.50
k_2	0.050	0.07	7.01
			$k_{tot} = 10.51$

Using the PASCO Capstone software to measure the oscillations, each trial commenced by setting the damper apparatus to a specified height by pressing it firmly against a known number of notecards stacked and placed on the track. The cart was then displaced by a uniform distance (10 cm) and released. As the cart oscillates, the first and second peaks' displacements were measured and solved for the *log decrement*, δ . This quantity measures the logarithmic ratio of the magnitudes of two successive peaks in an underdamped oscillation, and defined as

$$\delta = \ln\left(\frac{x_i}{x_{i+1}}\right). \text{ Then, the damping ratio can be found as } \zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}. \text{ This method will be}$$

repeated after adding a 250-g mass to the cart, and then once more with an additional 250-g mass, creating three different trials for each damper height. Iteration was conducted over several different damper heights, and the relationship between damper height and damping ratio was evaluated via plotting and statistical analysis. A line of best fit was used to fit the variation.



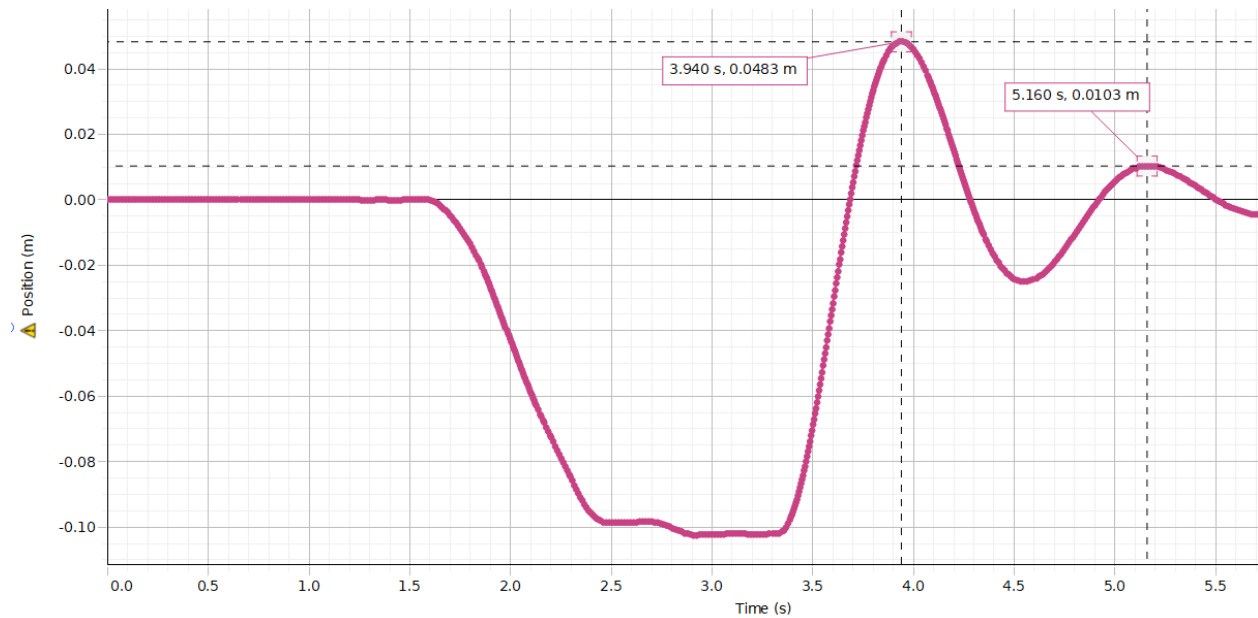


Figure 1. Example log decrement data collection at peaks of oscillation for two unique damping conditions.

The team selected 10 evenly spaced distances to prepare an appropriate model, varying by a reference width of five notecards. An additional data point was collected for the displacement of a single notecard to collect data as close to a height of zero as possible. The goodness of fit test performance will be evaluated based on its R^2 and p-values for significance. It will also be quantitatively compared to the hypothesized values to see if the uncertainty of the model falls within expectations.

Results + Analysis

Significant exponential relationships were present across each mass, with the damping increasing rapidly as the distance between the magnetic damper on the cart and the track was decreased. A visualization of the data collection is shown in the scatterplot below. It is

interesting to note that the different masses applied led to different damping ratios, which likely can be attributed to changes in the frictional forces between the cart and track.

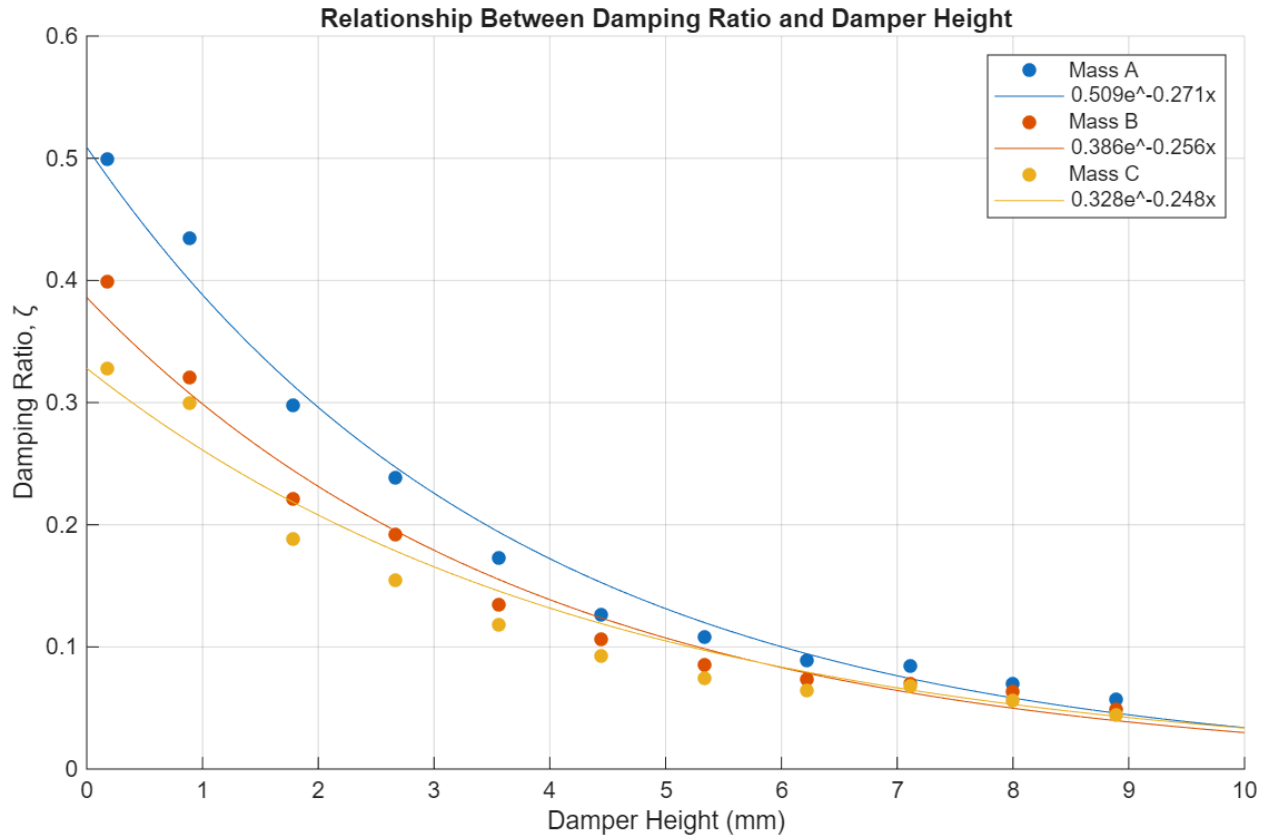


Figure 2. Scatterplot with fitted exponential regression lines of best fit for each trial conducted.

Each of these models were prescribed using a specific spring configuration with a spring constant $k = 10.5 \text{ N/m}$. To evaluate the robustness and durability of these models over varying conditions, an independent validation was conducted with an entirely unique spring-mass configuration. This verification process consisted of changing both springs and finding a predicted exponential decay based on the existing models, then conducting a root mean square error (RMSE) analysis to quantify the extremity of the residuals.

The springs used in the validation were only 62% as stiff as the ones used in the initial model, so the existing exponential was modified to accommodate for this. Because $\zeta = \frac{c}{2\sqrt{mk_{eq}}}$,

changing the existing equation for ζ based on the changed k_{eq} occurs via $\zeta_{val} = \zeta_{model} \sqrt{\frac{k_{model}}{k_{val}}}$.

The resulting exponential used in the independent validation was $\zeta_{val} = 0.6449e^{-0.0482h}$.

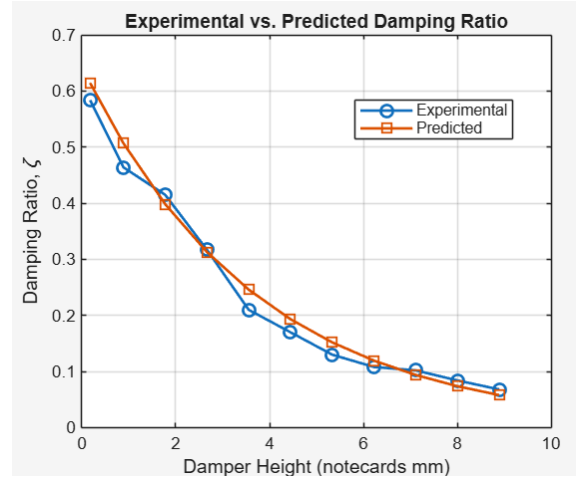


Figure 3. Experimental vs. predicted damping ratio at varying damper heights.

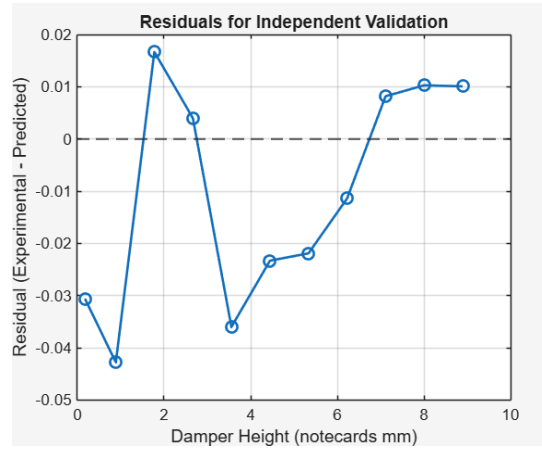


Figure 4. Residuals of experimental vs. predicted damping ratio at varying damper heights.

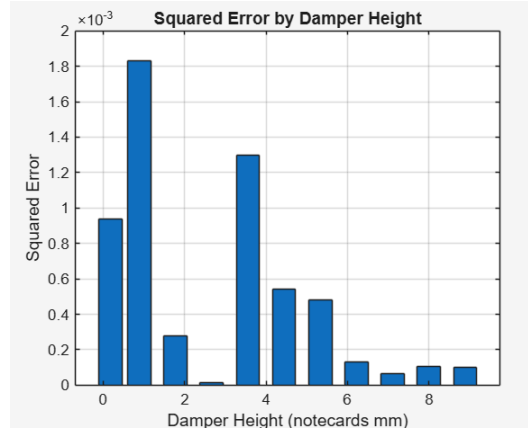


Figure 5. Squared error between predicted and experimental damping ratios by damper height.

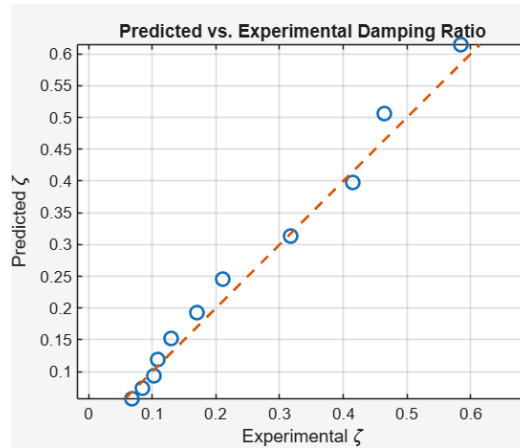


Figure 6. Predicted damping ratio compared to experimental damping ratio.

The experimentally measured damping ratios with the predicted values across different damper heights are compared in Figure 3. The two curves follow similar trends, indicating the model is reliable at capturing the relationship between damper height and damping ratio in this region. Figure 5 displays the squared error for each damper height, showing which experimental points contribute most to the overall model error. Large errors occur primarily at smaller damper heights, where the damping effect becomes stronger. This could also be the effect of microvariations in notecard height making more of a difference for small n .

Figure 6 shows a plot comparing predicted and experimental damping ratios. The closer the points are to the 1:1 line, the closer our predictive model aligns with the experimental data.

The root mean square error between the experimental damping ratios and predicted model values was calculated using the RMSE formula as follows:

$$\sqrt{\frac{1}{n} \sum (z_{exp} - z_{pred})^2}$$

The resulting RMSE was 0.0229, indicating that the model predictions deviate from the measured values by about 0.0229 damping ratio units on average, suggesting the fitted model reasonably represents the relationship between damper height and damping ratio over the testing range. By comparison, the root mean square error of each of the lines of best fit for each of the three masses were 0.0177, 0.0164, and 0.0175, respectively. Therefore, the residuals remained relatively accurate, although some variation in the model led it to be worse than each of the three best-fit regressions.

The evidence discovered here strongly suggests that the relationship between magnetic damping distance and force follows an exponential decline. However, this seems to conflict with the generally accepted idea that induced eddy currents weaken following an inverse cube pattern. Previous research is limited, but one source seems to suggest (Sodano & Bae, 2004) that, at close distances, the pattern can reveal itself as exponential. Further investigation into this underdeveloped area of research could include testing for similar relationships in other magnetic damping contexts or geometries to see if the relationship holds. Other potential areas for future expansion of this experiment include using tools to measure the magnitudes of the magnetic forces being induced by the system and repeating with alternative configurations to ensure the relationship remains consistent across a broad spectrum of spring and mass values.

Conclusion

This experiment investigated the effect of magnetic damper height on the damping ratio of an underdamped spring mass system experiencing eddy current damping. By measuring the decay of oscillation amplitudes and applying the logarithmic decrement method, the damping ratio was determined for multiple damper heights and masses. The results verified that this relationship can be approximated accurately at near distances through an exponential model by independently validating a predictive model on a previously untested system, producing a root mean square error of 0.0229. Impacts on the behavior of the system were notable as the masses changed, resulting in slightly greater damping than expected between masses. This could be attributed to increased frictional forces under a greater load mass load which wouldn't have been accounted for. Ultimately, the study was able to largely uphold the hypothesis while offering valuable insights into specific characteristics of how spring-mass systems behave when subjected to magnetic damping.

References

- K. Nagaya, H. Kojima, Y. Karube and H. Kibayashi, "Braking forces and damping coefficients of eddy current brakes consisting of cylindrical magnets and plate conductors of arbitrary shape," in IEEE Transactions on Magnetics, vol. 20, no. 6, pp. 2136-2145, November 1984, doi: 10.1109/TMAG.1984.1063570.
- Sodano, Henry & Bae, Jae-Sung. (2004). Eddy Current Damping in Structures. The Shock and Vibration Digest. 36. 469-478. 10.1177/0583102404048517.